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E-Learning Material

Fundamentals of Mathematics

(25VBTOC102)

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Module: 01

Matrices & Linear Algebra

Module Information

Field	Details
Course Title	Fundamentals of Mathematics
Course Code	25VBT0C102
Module Number	Module 1 of 5
Module Title	Matrices and Linear Algebra

Module Overview

Mathematics is the language of science, engineering, and computing. Among all mathematical tools available, matrices and linear algebra stand as one of the most powerful and widely applicable. From solving simultaneous equations in circuit analysis to powering modern machine learning models, the concepts you will learn in this module form a cornerstone of applied mathematics.

This module, Matrices and Linear Algebra, is your first deep dive into structured mathematical thinking at the undergraduate level. It builds upon basic algebraic skills and elevates them to handle multiple variables, large systems of equations, and systematic computation methods. The emphasis throughout is on understanding not just how to compute, but why each method works.

You will begin by studying matrices — rectangular arrays of numbers — and the complete algebra that governs them: how to add, subtract, multiply, and transpose them, and crucially, what it means for these operations to be valid or undefined. This arithmetic foundation is essential for everything that follows.

Determinants form the next major topic. Every square matrix has a scalar value associated with it called the determinant, which encodes deep information about the matrix — whether it has an inverse, whether a system of equations has a unique solution, and how the matrix scales areas and volumes geometrically. You will compute determinants systematically using cofactor expansion and learn the elegant properties that make determinants a powerful analytical tool.

Building on determinants, you will study the adjoint and inverse of a matrix, and the critical distinction between singular and non-singular matrices. These concepts directly enable you to solve systems of linear equations — the primary motivation for most of this module.

The module then presents four distinct methods for solving linear systems: Cramer's Rule (determinant-based), Gauss Elimination, Gauss-Jordan Elimination, and LU Decomposition. Each method has different computational characteristics and is suited to different problem sizes and types. You will also study the concept of matrix rank and echelon forms, which determine whether a system is consistent, inconsistent, or has infinitely many solutions.

Finally, the Gauss-Seidel iterative method introduces you to numerical computation — solving large systems not by exact algebraic steps, but by iteratively refining an approximation until the desired accuracy is achieved. This approach is fundamental in engineering simulations and data science applications.

Learning Objectives

By the end of this module, you will be able to:

- Recall and identify all standard types of matrices and apply the rules of matrix algebra including addition, multiplication, and transposition.
- Understand the definition and geometric significance of determinants and compute them systematically using cofactor expansion.
- Understand the concepts of adjoint, inverse, singularity, and the consistency of linear systems.
- Apply Cramer's Rule and Gauss Elimination methods to solve systems of linear equations.
- Identify the rank of a matrix and use LU decomposition and Gauss-Seidel method for solving linear systems.

Learning Outcomes

After completing this module, you will be able to:

- Classify matrices by type and perform all four fundamental matrix operations correctly, including conditions for validity.
- Compute the determinant of 2×2 and 3×3 matrices using cofactor expansion and apply determinant properties.
- Find the inverse of a non-singular matrix using the adjoint method and elementary row operations.
- Solve a system of linear equations using Cramer's Rule, Gauss Elimination, and Gauss-Jordan Elimination.
- Determine the rank of a matrix, reduce it to echelon form, and apply LU Decomposition and Gauss-Seidel iteration.

Table of Topics

Sub-Unit	Topic	Description
1.1	Introduction to Matrices	Definition, order, notation, and types of matrices
1.2	Algebra of Matrices	Addition, subtraction, scalar multiplication, matrix multiplication
1.3	Transpose of a Matrix	Definition, properties, symmetric and skew-symmetric matrices
1.4	Determinants	Definition, expansion of 2×2 and 3×3 determinants, Sarrus rule
1.5	Minors, Cofactors and Properties	Minor and cofactor definitions, eight key properties of determinants
1.6	Adjoint and Inverse of a Matrix	Cofactor matrix, adjoint, inverse formula A^{-1} = adj(A)/det(A)
1.7	Singular Matrices and Consistency	Singular vs non-singular, homogeneous and non-homogeneous systems
1.8	Cramer's Rule	Determinant-based solution for 2×2 and 3×3 linear systems
1.9	Matrix Rank, Echelon and Normal Form	Rank definition, REF, RREF, Rouché-Capelli theorem
1.10	Gauss Elimination and Gauss-Jordan	Forward elimination, back substitution, full RREF solution
1.11	Gauss-Seidel Iterative Method	Diagonal dominance, iterative computation, convergence
1.12	LU Decomposition	Doolittle method, forward/back substitution via L and U

1.1 Introduction to Matrices

A matrix is a rectangular arrangement of numbers, symbols, or expressions organised into rows and columns. The concept of a matrix is not merely an abstract algebraic curiosity — it is a fundamental data structure that underlies everything from the transformation of 3D graphics on your screen to the weights inside a neural network. Understanding matrices is, in a real sense, understanding how modern quantitative science organises and processes information.

The individual entries of a matrix are called elements or entries. By convention, a matrix is denoted by an uppercase letter such as A, B, or C, while its elements are denoted by the corresponding lowercase letter with two subscripts indicating the row and column positions.

Notation and Order

A matrix A with m rows and n columns is said to be of order m × n (read "m by n"). The element in the i-th row and j-th column is written as a_ij. The general form of a matrix of order m × n is written as:

General form: A = [a_ij] where i = 1, 2, ..., m and j = 1, 2, ..., n

For example, the 3 × 3 matrix A with elements written as a_11, a_12, a_13 in the first row, and so on:

A =

a_{11}	a_{12}	a_{13}
a_{21}	a_{22}	a_{23}
a_{31}	a_{32}	a_{33}

A specific 3 × 3 matrix with numerical values might look like:

A =

1	2	3
4	5	6
7	8	9

Types of Matrices

Matrices come in many special forms, each with unique properties that make them useful in different contexts. The following types are fundamental and appear throughout this course and beyond.

Type	Definition	Example / Property
Row Matrix	Has exactly one row (order 1 × n)	A = [3 -1 5] — order 1×3
Column Matrix	Has exactly one column (order m × 1)	B = [2 ; 0 ; 7] — order 3×1
Square Matrix	Number of rows = number of columns (order n × n)	Any 2×2, 3×3 etc. matrix
Diagonal Matrix	Square matrix; all non-diagonal elements are zero	diag(2, 5, 9)
Scalar Matrix	Diagonal matrix with all diagonal elements equal	diag(k, k, k) = k·I

Identity (Unit) Matrix	Diagonal matrix with all diagonal elements = 1; denoted I_n	I_3 = diag(1,1,1)
Zero (Null) Matrix	All elements are zero; denoted O or 0	O = [0 0; 0 0]
Upper Triangular	All elements below the main diagonal are zero	a_ij = 0 for i > j
Lower Triangular	All elements above the main diagonal are zero	a_ij = 0 for i < j
Symmetric Matrix	A^T = A (element a_ij = a_ji)	Transpose equals original
Skew-Symmetric	A^T = −A (a_ij = −a_ji; diagonal = 0)	Main diagonal is all zeros

Let us see explicit examples of the most important special matrices:

Identity Matrix I_3 =

1	0	0
0	1	0
0	0	1

Upper Triangular U =

2	3	1
0	5	4
0	0	7

Symmetric S =

4	2	1
2	6	3
1	3	8

Did You Know?

The identity matrix plays the same role in matrix algebra that the number 1 plays in ordinary arithmetic. Just as $1 \times a = a \times 1 = a$ for any number a, the identity matrix satisfies $A \cdot I = I \cdot A = A$ for any compatible square matrix A. This property makes it essential in computing inverses, eigenvalues, and transformations in computer graphics engines like those used in Unreal Engine and the animation software developed by Tata Elxsi.

1.2 Algebra of Matrices

Just as we define arithmetic operations for numbers, we can define analogous operations for matrices. However, matrix algebra has important differences from scalar arithmetic — the most notable being that multiplication is not generally commutative. Understanding these operations deeply is essential, as they form the computational backbone of all matrix-based methods in this course.

Matrix Equality

Two matrices A and B are equal (written $A = B$) if and only if they have the same order $m \times n$, and every corresponding element is equal: $a_{ij} = b_{ij}$ for all i and j. Matrices of different orders cannot be compared for equality.

Matrix Addition and Subtraction

Matrix addition is defined only when two matrices have the same order. If A and B are both $m \times n$ matrices, their sum $C = A + B$ is obtained by adding corresponding elements: $c_{ij} = a_{ij} + b_{ij}$. The result is also an $m \times n$ matrix. Subtraction follows the same element-wise rule: $c_{ij} = a_{ij} - b_{ij}$.

Properties of matrix addition (where A, B, C are matrices of the same order and O is the zero matrix):

- Commutativity: $A + B = B + A$
- Associativity: $(A + B) + C = A + (B + C)$
- Additive identity: $A + O = O + A = A$
- Additive inverse: $A + (-A) = O$

📝 Worked Example: Matrix Addition

Let A and B be 2×2 matrices. Compute A + B.

A =

3	1

−2	4

B =

1	5
6	−3

Step 1: Add corresponding elements row by row:

A + B =

3+1	1+5
(−2)+6	4+(−3)

A + B =

4	6
4	1

Scalar Multiplication

A matrix A can be multiplied by a scalar (an ordinary number) k. Each element of the matrix is multiplied by k: if B = kA, then b_ij = k · a_ij. This operation always produces a matrix of the same order.

Key properties (k, l are scalars, A, B are matrices of the same order):

- k(A + B) = kA + kB (distributive over matrix addition)
- (k + l)A = kA + lA (distributive over scalar addition)
- (kl)A = k(lA) (associativity with scalars)
- 1·A = A, 0·A = O (identity and zero scalar)

Matrix Multiplication

Matrix multiplication is the most significant and distinctive operation in matrix algebra. Unlike addition, which is element-wise, multiplication involves a row-by-column dot product operation. The product AB is defined only when the number of columns in A equals the number of rows in B.

If A is of order $m \times p$ and B is of order $p \times n$, then the product $C = AB$ is of order $m \times n$. The element in the i-th row and j-th column of C is computed as the dot product of the i-th row of A with the j-th column of B:

Element formula: $c_{ij} = \Sigma(k=1 \text{ to } p)\ a_{ik} \cdot b_{kj} = a_{i1} \cdot b_{1j} + a_{i2} \cdot b_{2j} + \ldots + a_{ip} \cdot b_{pj}$

Critical Properties of Matrix Multiplication:

- NOT commutative: $AB \neq BA$ in general — this is the most important difference from scalar arithmetic.
- Associative: $(AB)C = A(BC)$
- Distributive: $A(B + C) = AB + AC$ and $(A + B)C = AC + BC$
- Identity: $AI = IA = A$ for any square matrix A of matching order
- Zero: $AO = OA = O$ (but $AB = O$ does NOT necessarily imply $A = O$ or $B = O$)

📝 Worked Example: Matrix Multiplication (2×2)

Let A be 2×2 and B be 2×2. Compute C = AB.

A =

2	3
1	4

B =

5	1
2	6

Step 1: c_{11} = (Row 1 of A) · (Col 1 of B) = 2×5 + 3×2 = 10 + 6 = 16

Step 2: c_{12} = (Row 1 of A) · (Col 2 of B) = 2×1 + 3×6 = 2 + 18 = 20

Step 3: c_{21} = (Row 2 of A) · (Col 1 of B) = 1×5 + 4×2 = 5 + 8 = 13

Step 4: c_{22} = (Row 2 of A) · (Col 2 of B) = 1×1 + 4×6 = 1 + 24 = 25

C = AB =

16	20
13	25

Verification that BA ≠ AB:

Step 1: $(BA)_{11}$ = 5×2 + 1×1 = 11 ≠ 16 — confirming AB ≠ BA.

🔍 Analysing Misconceptions: Matrix Multiplication is Commutative

❌ MYTH: Since numbers commute under multiplication (ab = ba), matrices should also commute, so AB = BA always.

✅ REALITY: Matrix multiplication is NOT commutative. In general, AB ≠ BA. Even when both products are defined, reversing the order almost always gives a different result. This is one of the most fundamental distinctions between matrix algebra and scalar arithmetic. Special cases exist (e.g., any matrix commutes with the identity matrix, and any matrix commutes with its own powers), but these are exceptions, not the rule.

✓ Let's Check

Q.No	Type	Question
1	MCQ	If A is a 3×4 matrix and B is a 4×2 matrix, what is the order of the product AB?
2	MCQ	Which property does matrix addition satisfy that matrix multiplication does NOT?
3	Fill in Blank	Two matrices can be added only if they have the same ______.
4	Fill in Blank	In the product AB, element c_ij equals the dot product of the i-th ______ of A with the j-th column of B.
5	True/False	For matrices A and B of the same order, AB = BA is always true.
6	True/False	If A is a 2×3 matrix and B is a 3×4 matrix, the product AB exists and is of order 2×4.

Answers:

Q.No	Answer
1	c) 3×2 — the inner dimensions (4) must match; result has outer dimensions (3×2).
2	b) Commutativity — matrix addition satisfies A + B = B + A, but multiplication does not satisfy AB = BA.
3	order (same number of rows and columns)
4	row

5	False — AB = BA only in special cases (e.g., both are the identity matrix). In general AB ≠ BA.
6	True — A is 2×3, B is 3×4; inner dimensions (3) match, result is 2×4.

1.3 Transpose of a Matrix

The transpose of a matrix is one of the most frequently used operations in linear algebra. It is conceptually simple — swap rows and columns — yet its properties underlie important theorems about symmetric systems, orthogonality, and the relationship between a matrix and its inverse.

The transpose of a matrix A, denoted A^T (read "A transpose"), is obtained by interchanging the rows and columns of A. If A is of order m × n, then A^T is of order n × m. The element in position (i, j) of A becomes the element in position (j, i) of A^T:

Transpose definition: [A^T]_ij = [A]_ji

📝 Worked Example: Finding the Transpose

Find A^T for the following 2×3 matrix A.

A =

1	2	3
4	5	6

Step 1: A has order 2×3, so A^T will have order 3×2.

Step 2: Row 1 of A becomes Column 1 of A^T; Row 2 of A becomes Column 2 of A^T.

A^T =

1	4
2	5
3	6

Properties of the Transpose

The transpose satisfies the following important properties, which are used extensively in matrix algebra proofs and computations:

- (A^T)^T = A — Transposing twice returns the original matrix.
- (A + B)^T = A^T + B^T — Transpose distributes over addition.
- (kA)^T = k(A^T) — Scalar multiplication passes through transpose.
- (AB)^T = B^T · A^T — Note the reversal of order — this is critical!
- (ABC)^T = C^T B^T A^T — The reversal extends to products of any length.

Symmetric and Skew-Symmetric Matrices

A square matrix A is called symmetric if A^T = A, meaning a_ij = a_ji for all i, j. Symmetric matrices are highly significant in mathematics and physics — the covariance matrices in statistics, inertia tensors in mechanics, and Hessian matrices in optimisation are all symmetric.

A square matrix A is called skew-symmetric (or anti-symmetric) if A^T = −A, meaning a_ij = −a_ji for all i, j. An immediate consequence is that all diagonal elements of a skew-symmetric matrix must be zero, since a_ii = −a_ii implies a_ii = 0.

📝 Worked Example: Symmetric and Skew-Symmetric Example

Verify that S is symmetric and K is skew-symmetric.

S =

4	2	−1

2	7	3
−1	3	5

Step 1: Compute S^T by swapping rows and columns:

S^T =

4	2	−1
2	7	3
−1	3	5

Step 2: Since S^T = S, the matrix S is symmetric. ✓

K =

0	3	−2
−3	0	5
2	−5	0

Step 3: Compute K^T:

K^T =

0	−3	2
3	0	−5
−2	5	0

Step 4: Since K^T = −K, the matrix K is skew-symmetric. ✓ (All diagonal elements are 0.)

📌 **Note:** Any square matrix A can be uniquely expressed as the sum of a symmetric matrix and a skew-symmetric matrix: A = (1/2)(A + A^T) + (1/2)(A − A^T). The first part is symmetric; the second is skew-symmetric.

1.4 Determinants — Definition and Expansion

The determinant is one of the most profound concepts in linear algebra. To every square matrix, we can assign a single scalar value called its determinant, which encodes essential information about the matrix. Geometrically, the absolute value of the determinant represents the factor by which the matrix scales areas (in 2D) or volumes (in 3D). Algebraically, it tells us whether a matrix has an inverse, whether a linear system has a unique solution, and plays a central role in Cramer's Rule and the computation of eigenvalues.

The determinant of a square matrix A is denoted det(A), |A|, or Δ. Determinants are only defined for square matrices.

Determinant of a 1×1 Matrix

For a 1×1 matrix A = [a], the determinant is simply the element itself: det([a]) = a. This is the base case from which higher-order determinants are built.

Determinant of a 2×2 Matrix

For a 2×2 matrix, the determinant is computed as the difference of the products of the two diagonals — the principal diagonal minus the secondary diagonal:

A =

a	b
c	d

Determinant of 2×2: $\det(A) = |A| = ad - bc$

📝 Worked Example: Determinant of a 2×2 Matrix

Compute the determinant of the matrix:

A =

5	3
2	4

Step 1: Apply the formula: $\det(A) = a \cdot d - b \cdot c$

Step 2: $\det(A) = (5)(4) - (3)(2) = 20 - 6 = 14$

Therefore $\det(A) = 14$.

Determinant of a 3×3 Matrix — Cofactor Expansion

For a 3×3 matrix, the determinant is computed by expanding along any row or column using cofactors. Expanding along the first row is the most common approach:

A =

a_{11}	a_{12}	a_{13}
a_{21}	a_{22}	a_{23}
a_{31}	a_{32}	a_{33}

Expansion along Row 1: $\det(A) = a_{11} \cdot (a_{22} \cdot a_{33} - a_{23} \cdot a_{32}) - a_{12} \cdot (a_{21} \cdot a_{33} - a_{23} \cdot a_{31}) + a_{13} \cdot (a_{21} \cdot a_{32} - a_{22} \cdot a_{31})$

📝 Worked Example: Determinant of a 3×3 Matrix

Compute the determinant of the matrix:

A =

1	2	3
4	5	6
7	8	10

Step 1: Expand along Row 1. The signs follow the pattern: + − +.

Step 2: a_{11} = 1, Minor M_{11} = det([[5,6],[8,10]]) = (5)(10) − (6)(8) = 50 − 48 = 2

Step 3: a_{12} = 2, Minor M_{12} = det([[4,6],[7,10]]) = (4)(10) − (6)(7) = 40 − 42 = −2

Step 4: a_{13} = 3, Minor M_{13} = det([[4,5],[7,8]]) = (4)(8) − (5)(7) = 32 − 35 = −3

Step 5: det(A) = 1·(+2) − 2·(−2) + 3·(−3) = 2 + 4 − 9 = −3

Therefore det(A) = −3.

💡 Did You Know?

Determinants were first studied systematically by the Japanese mathematician Seki Takakazu around 1683 — before the European mathematicians Leibniz and Cramer. In modern computing, Intel's Math Kernel Library (MKL), used by numerical software at IISc Bangalore, TIFR, and HPC clusters like PARAM Rudra, computes determinants of matrices with thousands of rows using LU decomposition rather than cofactor expansion, since cofactor expansion is computationally expensive (factorial time complexity versus $O(n^3)$ for LU).

1.5 Minors, Cofactors, and Properties of Determinants

The concepts of minors and cofactors are the building blocks of determinant expansion. They allow us to reduce the computation of a higher-order determinant to a combination of lower-order determinants, enabling systematic and efficient calculation. Equally important are the properties of determinants — a set of rules that allow us to simplify computation dramatically before evaluating.

Minor of an Element

The minor M_{ij} of the element a_{ij} in a matrix A is the determinant of the $(n-1) \times (n-1)$ submatrix obtained by deleting the i-th row and j-th column from A. Every element in a square matrix has a corresponding minor.

Cofactor of an Element

The cofactor C_{ij} of the element a_{ij} is the minor M_{ij} with an appropriate sign determined by the position (i, j). The sign follows a checkerboard pattern — positive when (i+j) is even, negative when (i+j) is odd:

Cofactor definition: $C_{ij} = (-1)^{(i+j)} \cdot M_{ij}$

The sign pattern for a 3×3 matrix is:

Sign Pattern =

+	−	+
−	+	−
+	−	+

📝 Worked Example: Minors and Cofactors of a 3×3 Matrix

Find all minors and cofactors for Row 1 of:

A =

2	−1	3
1	4	−2
5	0	1

Step 1: Minor M_{11}: Delete Row 1, Col 1. Remaining submatrix:

4	−2
0	1

$M_{11} = (4)(1) - (-2)(0) = 4 - 0 = 4$. Cofactor $C_{11} = (-1)^{1+1} \cdot 4 = +4$

Step 2: Minor M_{12}: Delete Row 1, Col 2. Remaining submatrix:

1	−2
5	1

$M_{12} = (1)(1) - (-2)(5) = 1 + 10 = 11$. Cofactor $C_{12} = (-1)^{1+2} \cdot 11 = -11$

Step 3: Minor M_{13}: Delete Row 1, Col 3. Remaining submatrix:

1	4
5	0

$M_{13} = (1)(0) - (4)(5) = 0 - 20 = -20$. Cofactor $C_{13} = (-1)^{1+3} \cdot (-20) = -20$

Step 4: Verification: $\det(A) = a_{11} \cdot C_{11} + a_{12} \cdot C_{12} + a_{13} \cdot C_{13}$

$\det(A) = 2 \cdot (4) + (-1) \cdot (-11) + 3 \cdot (-20) = 8 + 11 - 60 = -41$

Properties of Determinants

These eight properties are extremely powerful tools. They allow you to simplify a determinant before evaluating it — often reducing a complex computation to a trivial one.

#	Property	Explanation
P1	$\det(A^T) = \det(A)$	The determinant is unchanged by transposition — rows and columns are interchangeable.
P2	Interchanging any two rows changes the sign of the determinant.	Useful for bringing a matrix to a standard form before evaluation.
P3	If any two rows are identical, $\det(A) = 0$.	A direct consequence of P2 (swapping identical rows changes sign but doesn't change the matrix).
P4	Multiplying any single row by a scalar k multiplies $\det(A)$ by k.	To factor out a scalar, pull it out of the determinant: $\det(kA) = k^n \cdot \det(A)$ for $n \times n$.
P5	If any row is entirely zero, $\det(A) = 0$.	Zero row means the matrix is singular.
P6	If one row is a scalar multiple of another, $\det(A) = 0$.	Linear dependence of rows implies $\det = 0$.
P7	Adding a scalar multiple of one row to another leaves $\det(A)$ unchanged.	This is the basis for Gauss elimination: row operations of this type preserve the determinant.

P8	det(AB) = det(A) · det(B) for square matrices A, B.	The determinant of a product equals the product of determinants.

🔍 Analysing Misconceptions: Row Operations Always Change the Determinant

❌ MYTH: Any row operation performed on a matrix changes its determinant.

✅ REALITY: Only specific row operations change the determinant. Adding a multiple of one row to another (the type used in Gauss elimination) leaves the determinant unchanged. Swapping two rows changes the sign of the determinant. Multiplying a row by scalar k multiplies the determinant by k. Knowing these rules precisely allows you to use row operations to simplify a matrix before computing its determinant.

1.6 Adjoint and Inverse of a Matrix

The inverse of a matrix is the matrix analogue of the reciprocal of a number. Just as dividing by a number n is the same as multiplying by 1/n, solving a matrix equation $AX = B$ often reduces to computing A^{-1} and then finding $X = A^{-1}B$. However, unlike numbers (where every nonzero number has a reciprocal), only certain matrices — the non-singular ones — have inverses. The adjoint method provides a complete, explicit formula for the inverse.

The Cofactor Matrix and Adjoint

The cofactor matrix (also called the matrix of cofactors) of A is the n×n matrix whose (i,j)-th element is the cofactor C_ij of A. The adjoint of A, written adj(A), is the transpose of the cofactor matrix:

Adjoint definition: adj(A) = [C_ij]^T

In other words, the element in position (i,j) of adj(A) is the cofactor of the element in position (j,i) of A — note the reversed indices due to transposition.

Inverse of a Matrix

The inverse of a square matrix A, denoted A^{-1}, exists if and only if $\det(A) \neq 0$. When the inverse exists, it is given by:

Inverse formula: $A^{-1} = \text{adj}(A) / \det(A) = (1/\det(A)) \cdot \text{adj}(A)$

This formula is known as the adjoint method or the classical adjoint method. It provides an explicit expression for every element of A^{-1}.

📝 Worked Example: Finding the Inverse of a 3×3 Matrix

Find the inverse of the matrix:

A =

2	1	0
1	3	1
0	1	2

Step 1: Compute det(A) by expanding along Row 1:

det(A) = 2·det([[3,1],[1,2]]) − 1·det([[1,1],[0,2]]) + 0

= 2·(6−1) − 1·(2−0) = 2·5 − 1·2 = 10 − 2 = 8

Since det(A) = 8 ≠ 0, the inverse exists.

Step 2: Find all 9 cofactors of A:

C_{11} = +det([[3,1],[1,2]]) = 6−1 = 5

C_{12} = −det([[1,1],[0,2]]) = −(2−0) = −2

C_{13} = +det([[1,3],[0,1]]) = 1−0 = 1

C_{21} = −det([[1,0],[1,2]]) = −(2−0) = −2

C_{22} = +det([[2,0],[0,2]]) = 4−0 = 4

C_{23} = −det([[2,1],[0,1]]) = −(2−0) = −2

C_{31} = +det([[1,0],[3,1]]) = 1−0 = 1

C_{32} = −det([[2,0],[1,1]]) = −(2−0) = −2

C_{33} = +det([[2,1],[1,3]]) = 6−1 = 5

Step 3: Write the cofactor matrix and transpose to get adj(A):

Cofactor Matrix =

5	−2	1
−2	4	−2
1	−2	5

adj(A) = [Cofactor Matrix]^T =

5	−2	1
−2	4	−2
1	−2	5

(Note: In this case adj(A) = cofactor matrix since the cofactor matrix is symmetric.)

Step 4: Compute A^{-1} = adj(A) / det(A) = adj(A) / 8:

A^{-1} = (1/8) ×

5	−2	1
−2	4	−2
1	−2	5

Step 5: Verify: $A \cdot A^{-1} = I_3$. (Product should give the 3×3 identity matrix.)

Note: For large matrices, computing the inverse via the adjoint method is computationally expensive (requires n^2 cofactors, each involving an $(n-1)\times(n-1)$ determinant). In practice, the inverse is computed using Gauss-Jordan Elimination by augmenting [A | I] and reducing to [I | A^{-1}]. This is covered in Sub-unit 1.10.

1.7 Singular and Non-Singular Matrices; Consistency of Linear Systems

Before studying methods for solving linear equations, it is essential to understand when a system can actually be solved. The concepts of singular and non-singular matrices directly answer this question: the determinant of the coefficient matrix is the key indicator of what kind of solution a system possesses.

Singular and Non-Singular Matrices

A square matrix A is called non-singular (or invertible) if its determinant is non-zero: $\det(A) \neq 0$. A non-singular matrix has a unique inverse A^{-1}. A square matrix A is called singular if its determinant is zero: $\det(A) = 0$. A singular matrix does not have an inverse.

Property	Non-Singular ($\det \neq 0$)	Singular ($\det = 0$)
Inverse	Exists: $A^{-1} = \text{adj}(A)/\det(A)$	Does not exist
Linear system AX = B	Unique solution: $X = A^{-1}B$	No solution or infinitely many
Rows/Columns	Linearly independent	At least one linear dependence
Rank	Full rank: $\text{rank}(A) = n$	$\text{rank}(A) < n$
Eigenvalues	None equal to zero	At least one eigenvalue = 0

Linear Systems — Formulation

A system of m linear equations in n unknowns $x_1, x_2, \ldots, x_n$ can be written compactly in matrix form. The system:

$$a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n = b_2$$

$$\vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n = b_m$$

is written as $AX = B$, where A is the $m \times n$ coefficient matrix, X is the $n \times 1$ column vector of unknowns, and B is the $m \times 1$ column vector of constants.

Homogeneous and Non-Homogeneous Systems

A system $AX = B$ is called homogeneous if $B = 0$ (all right-hand sides are zero) and non-homogeneous if $B \neq 0$. A homogeneous system $AX = 0$ always has the trivial solution $X = 0$. It has non-trivial solutions (solutions other than $X = 0$) if and only if $\det(A) = 0$ — i.e., if and only if A is singular.

Consistency — The Rouché-Capelli Theorem

The Rouché-Capelli theorem (also called the Rouché-Fontené or Kronecker-Capelli theorem) provides a complete criterion for the consistency of a linear system, using the concept of matrix rank (detailed in Sub-unit 1.9):

- The system $AX = B$ is consistent (has at least one solution) if and only if $\text{rank}(A) = \text{rank}([A|B])$.

- If the system is consistent and rank(A) = rank([A|B]) = n (the number of unknowns), then there is a unique solution.
- If the system is consistent and rank(A) = rank([A|B]) = r < n, then there are infinitely many solutions with (n − r) free parameters.
- If rank(A) ≠ rank([A|B]), the system is inconsistent (no solution exists).

🔍 Analysing Misconceptions: det(A) = 0 Means No Solution Exists

✕ MYTH: If the determinant of the coefficient matrix is zero, the system of equations has no solution.

✓ REALITY: A zero determinant means either no solution (inconsistent system) OR infinitely many solutions — not necessarily no solution. Which case applies depends on whether rank(A) = rank([A|B]) or not. You must check the augmented matrix [A|B] to distinguish between the two. For example, the system $x + y = 1$, $2x + 2y = 2$ has det = 0 but infinitely many solutions (not no solution).

1.8 Cramer's Rule

Cramer's Rule is an elegant theorem that expresses the solution of a system of linear equations directly in terms of determinants. Named after the Swiss mathematician Gabriel Cramer (1704–1752), this rule provides a closed-form formula for each unknown in a system, making it particularly transparent conceptually, even though for large systems, the Gauss methods are computationally more efficient.

Cramer's Rule applies to a square system $AX = B$ where A is an $n \times n$ non-singular matrix ($\det(A) \neq 0$). In such a system, there is exactly one solution, and that solution is given by:

Cramer's Rule: $x_i = D_i / D$ where $D = \det(A)$ and $D_i = \det(A_i)$

Here, A_i is the matrix formed by replacing the i-th column of A with the column vector B. That is, D_i is the determinant of the matrix obtained by substituting B in place of column i of A.

Cramer's Rule for a 2×2 System

For the system: $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$, we have:

2×2 system: $D = a_1b_2 - a_2b_1$, $D_1 = c_1b_2 - c_2b_1$, $D_2 = a_1c_2 - a_2c_1$

Solution: $x = D_1/D$, $y = D_2/D$

📝 Worked Example: Cramer's Rule — 2×2 System

Solve the system: $2x + 3y = 8$ and $x - y = 1$

Step 1: Write the coefficient matrix A and compute D = det(A):

A =

2	3
1	−1

$D = \det(A) = (2)(-1) - (3)(1) = -2 - 3 = -5$

Step 2: Form D_1 by replacing Column 1 with B = [8; 1]:

A_1 =

8	3
1	−1

$D_1 = (8)(-1) - (3)(1) = -8 - 3 = -11$

Step 3: Form D_2 by replacing Column 2 with B = [8; 1]:

A_2 =

2	8
1	1

$D_2 = (2)(1) - (8)(1) = 2 - 8 = -6$

Step 4: Apply Cramer's Rule:

$x = D_1/D = -11/(-5) = 11/5 = 2.2$

$y = D_2/D = -6/(-5) = 6/5 = 1.2$

Solution: $x = 11/5$, $y = 6/5$. Verify: $2(11/5) + 3(6/5) = 22/5 + 18/5 = 40/5 = 8$ ✓

📝 Worked Example: Cramer's Rule — 3×3 System

Solve: $x + y + z = 6$, $2x - y + z = 3$, $x + 2y - z = 2$

Step 1: Set up and compute D:

A =

1	1	1
2	−1	1
1	2	−1

$D = 1[(-1)(-1)-(1)(2)] - 1[(2)(-1)-(1)(1)] + 1[(2)(2)-(-1)(1)]$

$= 1[1-2] - 1[-2-1] + 1[4+1] = 1(-1) - 1(-3) + 1(5) = -1 + 3 + 5 = 7$

Step 2: Compute D_1 (replace Col 1 with [6;3;2]):

A_1 =

6	1	1

3	−1	1
2	2	−1

$D_1 = 6(1-2) - 1(-3-2) + 1(6+2) = 6(-1) - 1(-5) + 1(8) = -6 + 5 + 8 = 7$

Step 3: Compute D_2 (replace Col 2 with [6;3;2]):

A_2 =

1	6	1
2	3	1
1	2	−1

$D_2 = 1(-3-2) - 6(-2-1) + 1(4-3) = 1(-5) - 6(-3) + 1(1) = -5 + 18 + 1 = 14$

Step 4: Compute D_3 (replace Col 3 with [6;3;2]):

A_3 =

1	1	6
2	−1	3
1	2	2

$D_3 = 1(-2-6) - 1(4-3) + 6(4+1) = 1(-8) - 1(1) + 6(5) = -8 - 1 + 30 = 21$

Step 5: Apply Cramer's Rule: $x = D_1/D = 7/7 = 1$, $y = D_2/D = 14/7 = 2$, $z = D_3/D = 21/7 = 3$

Solution: $x = 1$, $y = 2$, $z = 3$. ✓

✓ Let's Check

Q.No	Type	Question
1	MCQ	Cramer's Rule can be applied when the coefficient matrix is:
2	MCQ	For a 3×3 system AX = B, how is the matrix A_2 formed for finding x_2?
3	Fill in Blank	In Cramer's Rule, $x_1 = D_1/D$ where D = ______ and D_1 = det of A with column 1 replaced by B.
4	Fill in Blank	Cramer's Rule gives a ______ solution when $\det(A) \neq 0$.
5	True/False	Cramer's Rule can be used even when the coefficient matrix is singular.
6	True/False	For a 2×2 system where $D = -5$, $D_1 = 10$, the value of $x_1 = -2$.

Answers:

Q.No	Answer
1	b) Non-singular (det(A) ≠ 0) — Cramer's Rule requires a non-zero determinant.
2	a) Replace the 2nd column of A with the column vector B.
3	det(A)
4	unique
5	False — Cramer's Rule applies only when det(A) ≠ 0. If det(A) = 0, it cannot be used.
6	True — $x_1 = D_1/D = 10/(-5) = -2$.

1.9 Matrix Rank, Echelon Form, and Normal Form

The rank of a matrix is one of the most important numerical invariants associated with a matrix. It tells us, in precise terms, the "effective dimension" of the information contained in the matrix — how many rows (or columns) are truly independent of each other. The rank is the bridge between the abstract algebra of matrices and the concrete question of how many solutions a linear system has.

Rank of a Matrix

The rank of a matrix A, written rank(A) or ρ(A), is defined as the maximum number of linearly independent rows (or equivalently, the maximum number of linearly independent columns) in A. Equivalently, the rank equals the order of the largest non-zero minor in A.

Important facts about rank:

- For an $m \times n$ matrix, $\text{rank}(A) \le \min(m, n)$.
- $\text{rank}(A) = 0$ if and only if A is the zero matrix.
- For a square $n \times n$ matrix: $\text{rank}(A) = n \Leftrightarrow \det(A) \neq 0 \Leftrightarrow$ A is non-singular.
- Row operations (of any type) do not change the rank of a matrix.
- $\text{rank}(A^T) = \text{rank}(A)$.

Row Echelon Form (REF)

A matrix is in Row Echelon Form if:

- All zero rows are at the bottom of the matrix.
- The leading entry (first non-zero element, called the pivot) of each non-zero row is to the right of the leading entry of the row above it.
- All entries below a pivot in the same column are zero.

The number of non-zero rows in the REF of a matrix equals its rank. This gives us the most practical method for computing rank.

Reduced Row Echelon Form (RREF)

A matrix is in Reduced Row Echelon Form (also called Row Canonical Form) if it satisfies all REF conditions plus:

- Every pivot is equal to 1.
- All entries above a pivot are also zero (not just below).

Worked Example: Finding Rank by Row Reduction

Find the rank of the matrix:

A =

1	2	3	4
2	4	6	8
3	6	10	14

Step 1: $R_2 \to R_2 - 2R_1$ (subtract 2 times Row 1 from Row 2):

1	2	3	4
0	0	0	0
3	6	10	14

Step 2: $R_3 \to R_3 - 3R_1$ (subtract 3 times Row 1 from Row 3):

1	2	3	4
0	0	0	0
0	0	1	2

Step 3: Swap R_2 and R_3 to bring the non-zero row up (standard REF form):

REF(A) =

1	2	3	4

0	0	1	2
0	0	0	0

There are 2 non-zero rows. Therefore rank(A) = 2.

Note: The zero row means the rows of A were not all independent — Row 2 was a multiple of Row 1.

Normal Form of a Matrix

By applying both row and column operations (elementary operations), any matrix A of order m × n and rank r can be reduced to a canonical form called the Normal Form or Hermite Normal Form:

Normal form: N = [I_r | O] (top: I_r with zeros; bottom: all zeros)

where I_r is the r × r identity matrix. This normal form immediately reveals the rank as r.

1.10 Gauss Elimination and Gauss-Jordan Elimination

Gauss Elimination is the most systematic and widely used method for solving linear systems. Unlike Cramer's Rule which requires determinant computations that scale poorly, Gauss Elimination uses only simple row operations and has a computational cost that grows as $O(n^3)$ — making it practical for systems with hundreds or thousands of unknowns. Every major numerical library in the world, from MATLAB to Python's NumPy, uses variants of Gauss Elimination as its primary linear solver.

Elementary Row Operations

Gauss Elimination uses three types of elementary row operations, all of which preserve the solution set of the system:

- Type 1 — Interchange: Swap two rows. Notation: $R_i \leftrightarrow R_j$
- Type 2 — Scaling: Multiply a row by a non-zero scalar k. Notation: $R_i \to kR_i$
- Type 3 — Row addition: Add a scalar multiple of one row to another. Notation: $R_i \to R_i + kR_j$

Gauss Elimination — Forward Elimination + Back Substitution

Gauss Elimination proceeds in two phases. In the forward elimination phase, the augmented matrix [A|B] is reduced to upper triangular (Row Echelon) form using Type 3 row operations (and Type 1 if needed to handle zero pivots). In the back substitution phase, the unknowns are solved one by one starting from the last equation.

📝 Worked Example: Gauss Elimination — 3×3 System

Solve the system: $2x + y - z = 8$, $-3x - y + 2z = -11$, $-2x + y + 2z = -3$

Step 1: Write the augmented matrix [A|B]:

[A|B] =

2	1	−1	|	8
−3	−1	2	|	−11
−2	1	2	|	−3

Step 2: Eliminate the first column entries below the pivot (2 in position R_1):

$R_2 \to R_2 + (3/2)R_1$: $[-3+(3/2)\cdot 2, -1+(3/2)\cdot 1, 2+(3/2)(-1), -11+(3/2)\cdot 8] = [0, 1/2, 1/2, 1]$

$R_3 \to R_3 + (1)R_1$: $[-2+2, 1+1, 2+(-1), -3+8] = [0, 2, 1, 5]$

After Step 2:

2	1	−1	|	8
0	1/2	1/2	|	1
0	2	1	|	5

Step 3: Eliminate the second column entry below the second pivot (1/2 in R_2):

$R_3 \to R_3 - 4R_2$: $[0, 2-4\cdot(1/2), 1-4\cdot(1/2), 5-4\cdot 1] = [0, 0, -1, 1]$

Upper Triangular:

2	1	−1	|	8
0	1/2	1/2	|	1

0	0	−1	|	1

Step 4: Back substitution:

From Row 3: $-z = 1 \implies z = -1$

From Row 2: $(1/2)y + (1/2)(-1) = 1 \implies (1/2)y = 3/2 \implies y = 3$

From Row 1: $2x + (3) - (-1) = 8 \implies 2x + 4 = 8 \implies x = 2$

Solution: $x = 2$, $y = 3$, $z = -1$. ✓

Gauss-Jordan Elimination

Gauss-Jordan Elimination extends Gauss Elimination by continuing the row operations until the matrix is in Reduced Row Echelon Form (RREF), rather than stopping at upper triangular form. The key difference is that in Gauss-Jordan, we also eliminate entries above each pivot — not just below. The result directly gives the solution without any back substitution.

Gauss-Jordan is also used to find the inverse of a matrix by working with the augmented matrix [A|I] and reducing the left side to the identity matrix: the right side becomes A^{-1}.

📝 Worked Example: Gauss-Jordan — Finding A^{-1}

Find the inverse of A using Gauss-Jordan on the augmented matrix [A|I]:

A =

1	0	1
2	1	3
1	1	1

Step 1: Form the augmented matrix $[A|I_3]$:

[A|I] =

1	0	1	|	1	0	0
2	1	3	|	0	1	0
1	1	1	|	0	0	1

Step 2: $R_2 \to R_2 - 2R_1$ and $R_3 \to R_3 - R_1$:

1	0	1	|	1	0	0
0	1	1	|	−2	1	0
0	1	0	|	−1	0	1

Step 3: $R_3 \to R_3 - R_2$:

1	0	1	|	1	0	0
0	1	1	|	−2	1	0
0	0	−1	|	1	−1	1

Step 4: $R_3 \to -R_3$:

1	0	1	|	1	0	0
0	1	1	|	−2	1	0

0	0	1	|	−1	1	−1

Step 5: $R_2 \rightarrow R_2 - R_3$ and $R_1 \rightarrow R_1 - R_3$:

$[I|A^{-1}]$ =

1	0	0	|	2	−1	1
0	1	0	|	−1	0	1
0	0	1	|	−1	1	−1

Therefore A^{-1} = [[2,−1,1], [−1,0,1], [−1,1,−1]]

1.11 Gauss-Seidel Iterative Method

The methods studied so far — Gauss Elimination, Gauss-Jordan, and Cramer's Rule — are all direct methods: they compute the exact solution in a finite, predetermined number of steps. Iterative methods take a fundamentally different approach: they start with an initial guess and repeatedly improve it, converging toward the true solution. The Gauss-Seidel method, developed by Carl Friedrich Gauss and Philipp Ludwig von Seidel, is one of the most widely used iterative solvers, especially for large sparse systems that arise in engineering simulations and finite element analysis.

The key advantage of iterative methods is efficiency for large systems where the coefficient matrix has many zero entries (sparse matrices). Direct methods require $O(n^3)$ operations regardless of sparsity; iterative methods can exploit sparsity to run much faster.

Diagonal Dominance — Condition for Convergence

The Gauss-Seidel method is guaranteed to converge if the coefficient matrix A is diagonally dominant. A matrix A is strictly diagonally dominant if, for every row i, the absolute value of the diagonal element exceeds the sum of absolute values of all other elements in that row:

Diagonal dominance condition: $|a_{ii}| > \Sigma_{(j\neq i)} |a_{ij}|$ for all i

If the matrix is not diagonally dominant, the equations should be rearranged (if possible) to achieve diagonal dominance before applying Gauss-Seidel.

The Gauss-Seidel Algorithm

Given the system AX = B, we isolate each unknown from its own equation. For a 3×3 system with equations $a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$, etc., the iterative formulas are:

Update x_1: $x_1^{(k+1)} = (1/a_{11}) \cdot [\, b_1 - a_{12} \cdot x_2^{(k)} - a_{13} \cdot x_3^{(k)} \,]$

Update x_2: $x_2^{(k+1)} = (1/a_{22}) \cdot [\, b_2 - a_{21} \cdot x_1^{(k+1)} - a_{23} \cdot x_3^{(k)} \,]$

Update x_3: $x_3^{(k+1)} = (1/a_{33}) \cdot [\, b_3 - a_{31} \cdot x_1^{(k+1)} - a_{32} \cdot x_2^{(k+1)} \,]$

The critical feature of Gauss-Seidel (distinguishing it from the older Jacobi method) is that it immediately uses the most recently computed values. When computing x_2, it uses the already-updated $x_1^{(k+1)}$, not the old $x_1^{(k)}$. This "use the latest available value" strategy is what makes Gauss-Seidel converge faster than Jacobi.

Worked Example: Gauss-Seidel — 3 Iterations

Solve the system using Gauss-Seidel (start from $x_1=x_2=x_3=0$, perform 3 iterations):

$10x_1 + x_2 + x_3 = 12$

$2x_1 + 10x_2 + x_3 = 13$

$x_1 + x_2 + 10x_3 = 14$

(Note: Diagonal dominance holds since $|10| > |1|+|1| = 2$ for each row.)

Step 1: Isolate each variable:

$x_1 = (1/10)(12 - x_2 - x_3)$

$x_2 = (1/10)(13 - 2x_1 - x_3)$

$x_3 = (1/10)(14 - x_1 - x_2)$

Step 2: Iteration 1 — Start with $x_1=0$, $x_2=0$, $x_3=0$:

$x_1 = (1/10)(12 - 0 - 0) = 1.2$

$x_2 = (1/10)(13 - 2(1.2) - 0) = (1/10)(10.6) = 1.06$

$x_3 = (1/10)(14 - 1.2 - 1.06) = (1/10)(11.74) = 1.174$

Step 3: Iteration 2 — Use values from Iteration 1:

$x_1 = (1/10)(12 - 1.06 - 1.174) = (1/10)(9.766) = 0.9766$

$x_2 = (1/10)(13 - 2(0.9766) - 1.174) = (1/10)(9.8728) = 0.9873$

$x_3 = (1/10)(14 - 0.9766 - 0.9873) = (1/10)(12.0361) = 1.2036$

Step 4: Iteration 3:

$x_1 = (1/10)(12 - 0.9873 - 1.2036) = (1/10)(9.8091) = 0.9809$

$x_2 = (1/10)(13 - 2(0.9809) - 1.2036) = (1/10)(9.8346) = 0.9835$

$x_3 = (1/10)(14 - 0.9809 - 0.9835) = (1/10)(12.0356) = 1.0036$

Converging toward the exact solution: $x_1 = 1$, $x_2 = 1$, $x_3 = 1$.

1.12 LU Decomposition

LU Decomposition is a matrix factorisation technique that expresses a square matrix A as the product of a Lower triangular matrix L and an Upper triangular matrix U: A = LU. This factorisation is extraordinarily useful because once A is decomposed into L and U, solving the system AX = B reduces to two simple triangular solves that are computationally cheap. Moreover, if you need to solve the same system AX = B for multiple different right-hand sides B, you only perform the decomposition once — a significant computational saving.

LU Decomposition is used by virtually every major scientific computing library. NumPy's numpy.linalg.solve, MATLAB's backslash operator, and R's solve() all internally use LU decomposition (via the LAPACK library). At institutions like IIT Bombay, IIT Madras, and TIFR, finite element simulations for structural analysis and computational fluid dynamics rely on LU decomposition to solve systems with millions of unknowns.

Doolittle's LU Decomposition

In Doolittle's method, the lower triangular matrix L has all diagonal elements equal to 1 (it is "unit lower triangular"), while U is a standard upper triangular matrix. The elements of L and U are computed systematically using the following formulas:

For each k from 1 to n:

Elements of U: $u_{kj} = a_{kj} - \Sigma(m=1 \text{ to } k-1)\, l_{km} \cdot u_{mj}$ for $j = k, k+1, \ldots, n$

Elements of L: $l_{ik} = (1/u_{kk}) \cdot [a_{ik} - \Sigma(m=1 \text{ to } k-1)\, l_{im} \cdot u_{mk}]$ for $i = k+1, \ldots, n$

Solving AX = B via LU Decomposition

Once A = LU is known, the system AX = B is solved in two steps:

- Forward substitution: Solve LY = B for Y (easy since L is lower triangular).
- Back substitution: Solve UX = Y for X (easy since U is upper triangular).

📝 Worked Example: LU Decomposition — 3×3 Matrix

Decompose A into LU, then solve AX = B where B = [1; 2; 3]:

A =

2	1	1
4	3	3
8	7	9

Step 1: Compute Row 1 of U (k=1): U[1] = Row 1 of A since L diagonal = 1

$u_{11} = 2, \quad u_{12} = 1, \quad u_{13} = 1$

Step 2: Compute Column 1 of L below diagonal:

$l_{21} = a_{21}/u_{11} = 4/2 = 2$

$l_{31} = a_{31}/u_{11} = 8/2 = 4$

Step 3: Compute Row 2 of U (k=2):

$u_{22} = a_{22} - l_{21} \cdot u_{12} = 3 - (2)(1) = 1$

$u_{23} = a_{23} - l_{21} \cdot u_{13} = 3 - (2)(1) = 1$

Step 4: Compute Column 2 of L below diagonal:

$l_{32} = (a_{32} - l_{31} \cdot u_{12})/u_{22} = (7 - 4 \cdot 1)/1 = 3$

Step 5: Compute Row 3 of U (k=3):

$u_{33} = a_{33} - l_{31} \cdot u_{13} - l_{32} \cdot u_{23} = 9 - 4 \cdot 1 - 3 \cdot 1 = 2$

Step 6: Result — the decomposition A = LU:

L =

1	0	0
2	1	0
4	3	1

U =

2	1	1
0	1	1
0	0	2

Step 7: Solve LY = B = [1; 2; 3] by forward substitution:

$1 \cdot y_1 = 1 \implies y_1 = 1$

$2y_1 + y_2 = 2 \implies y_2 = 2 - 2 = 0$

$4y_1 + 3y_2 + y_3 = 3 \implies y_3 = 3 - 4 - 0 = -1$

Step 8: Solve UX = Y = [1; 0; −1] by back substitution:

$2x_3 = -1 \implies x_3 = -1/2$

$x_2 + x_3 = 0 \implies x_2 = 1/2$

$2x_1 + x_2 + x_3 = 1 \implies 2x_1 = 1 - 1/2 + 1/2 = 1 \implies x_1 = 1/2$

Solution: $x_1 = 1/2$, $x_2 = 1/2$, $x_3 = -1/2$.

Let's Check

Q.No	Type	Question
1	MCQ	In LU Decomposition using Doolittle's method, which matrix has all diagonal elements equal to 1?
2	MCQ	After computing A = LU, what is the first step in solving AX = B?
3	Fill in Blank	The Gauss-Seidel method converges when the coefficient matrix is strictly ______.
4	Fill in Blank	In the Gauss-Jordan method, the augmented matrix [A|I] is reduced so that the left portion becomes ______ to find A^{-1}.
5	True/False	The Gauss-Seidel method immediately uses the updated values of variables as soon as they are computed in the current iteration.
6	True/False	LU Decomposition is most useful when you need to solve the same matrix system for many different right-hand side vectors B.

Answers:

Q.No	Answer
1	a) L (Lower triangular matrix) — in Doolittle's method, all l_ii = 1.
2	b) Solve LY = B by forward substitution — this gives the intermediate vector Y.
3	diagonally dominant
4	the identity matrix (I)
5	True — this distinguishes Gauss-Seidel from Jacobi (which uses only old values).
6	True — you perform the LU factorisation once ($O(n^3)$) and then each new B requires only $O(n^2)$ forward/back substitution.

📘 Case Study: ISRO's Mars Orbiter Mission (Mangalyaan) — Linear System Solving in Orbital Mechanics

Background:

India's Mars Orbiter Mission (MOM), popularly known as Mangalyaan, was launched by ISRO on 5 November 2013 and successfully entered Mars orbit on 24 September 2014 — making India the first country to achieve this on its first attempt, at a fraction of the cost of previous missions ($74 million). Behind this remarkable achievement was sophisticated mathematical computation, much of which relied directly on the linear algebra you have studied in this module.

Mathematical Challenge:

At every stage of the mission — trajectory planning, attitude control, orbital insertion — the flight dynamics team at ISRO's ISTRAC (ISRO Telemetry, Tracking and Command Network) in Bengaluru needed to solve large systems of linear equations. Orbital mechanics is governed by differential equations whose discretised numerical solution reduces to matrix systems of the form AX = B at each time step.

For example, to compute the precise thrust vector needed for the Trans-Mars Injection (TMI) burn, engineers solved a system of 12 equations in 12 unknowns representing the spacecraft's position, velocity, and orientation in three dimensions, plus fuel consumption constraints. The coefficient matrix A encoded the physical constraints of the spacecraft dynamics, and B encoded the desired final state (specific orbital parameters at Mars).

Matrix Methods Applied:

- LU Decomposition was used as the core linear solver, allowing the same orbital model (same A) to be solved for many different right-hand side vectors B as mission parameters were refined — exactly the scenario where LU excels.

- Gauss Elimination with partial pivoting was used to solve attitude control equations in real time, computing the corrections needed to keep the spacecraft oriented correctly during the 300-day cruise phase.
- Matrix rank computation (via Gauss-Jordan) was used to verify that sensor measurement equations were independent — a rank-deficient system of measurements would have indicated redundant or conflicting sensor data.
- Gauss-Seidel iteration was used in thermal modelling of the spacecraft body — solving the heat diffusion equations (which produce large sparse matrices) governing the temperature distribution across solar panels.

Significance:

The success of Mangalyaan demonstrated that India's scientific and mathematical capabilities could match global standards at a fraction of the cost. The mathematical tools covered in this module — determinants, matrix inversion, Gauss elimination, LU decomposition — are not academic exercises but the actual computational machinery operating inside India's most celebrated space mission. ISRO continues to use advanced linear algebra in ongoing missions including Chandrayaan-3 (2023 Moon landing) and the Aditya-L1 solar observatory (2023).

Module Summary

This module has provided a comprehensive foundation in matrices and linear algebra — one of the most practically powerful areas of undergraduate mathematics. The journey began with the fundamental concept of a matrix: a structured rectangular array with a complete algebra governing its operations.

You learned that matrix arithmetic is analogous to but importantly different from scalar arithmetic. Addition and scalar multiplication behave exactly as expected. Matrix multiplication, however, is not commutative — this single fact distinguishes matrix algebra from everything encountered before and has profound consequences in all applications.

The determinant emerged as the key scalar invariant of a square matrix — simultaneously a computational tool, a geometric measure, and a criterion for invertibility. The eight properties of determinants allow dramatic simplification of computation, and the systematic cofactor expansion provides a complete algorithm for evaluation.

Key conceptual landmarks in this module:

- Every square matrix has a determinant; only non-singular matrices (det ≠ 0) have inverses.
- The inverse $A^{-1} = \text{adj}(A)/\det(A)$ exists if and only if $\det(A) \neq 0$.
- The rank of a matrix quantifies the number of linearly independent rows/columns and directly determines the solution type of an associated linear system via the Rouché-Capelli theorem.
- Cramer's Rule gives an elegant explicit formula for each unknown but is computationally expensive for large systems.

- Gauss Elimination (forward elimination + back substitution) and Gauss-Jordan (full reduction to RREF) are the standard direct solution methods — efficient, systematic, and universally applicable.
- Gauss-Seidel iteration converges for diagonally dominant systems and is particularly powerful for large sparse matrices in engineering applications.
- LU Decomposition factorises $A = LU$ once, then solves any number of systems $AX = B$ cheaply via two triangular solves.

These methods form the computational engine of applied mathematics in science and engineering. Whether you are analysing a circuit, simulating fluid dynamics, training a machine learning model, or computing a satellite trajectory, the matrix operations and linear system solvers in this module are at work.

Glossary

Term	Definition
Matrix	A rectangular array of numbers arranged in rows and columns, denoted by an uppercase letter. Its order ($m \times n$) specifies the number of rows m and columns n.
Determinant	A scalar value computed from a square matrix that encodes whether the matrix is invertible, and geometrically represents the scaling factor of the transformation.
Minor (M_ij)	The determinant of the submatrix obtained by deleting row i and column j from the original matrix.
Cofactor (C_ij)	The minor M_ij with sign (−1)^(i+j) applied: C_ij = (−1)^(i+j)·M_ij.
Adjoint (adj A)	The transpose of the cofactor matrix of A. Used in the formula for the inverse: A^{-1} = adj(A)/det(A).
Singular Matrix	A square matrix with determinant equal to zero; it has no inverse.
Non-Singular Matrix	A square matrix with non-zero determinant; it possesses a unique inverse A^{-1}.
Rank	The maximum number of linearly independent rows (or columns) in a matrix; equals the order of the largest non-zero minor.
Row Echelon Form	A matrix form where all zero rows are at the bottom, and each pivot is to the right of the pivot in the row above.
LU Decomposition	The factorisation A = LU where L is unit lower triangular and U is upper triangular; used to efficiently solve systems AX = B.

Self Assessment Questions

Section A: Multiple Choice Questions (MCQ)

Choose the most appropriate answer for each question.

1. A matrix of order m × n has:

   a) m columns and n rows

   b) m rows and n columns

   c) m × n diagonal elements

   d) Equal rows and columns always

2. Which of the following is a scalar matrix?

   a) [[1,0],[0,2]]

   b) [[3,0],[0,3]]

   c) [[0,1],[1,0]]

   d) [[2,1],[0,2]]

3. If A is a 2×3 matrix and B is a 3×4 matrix, the product AB has order:

   a) 3×3

   b) 2×4

   c) 3×4

   d) 2×3

4. Which property does NOT hold for matrix multiplication?

   a) Associativity: (AB)C = A(BC)

   b) Commutativity: AB = BA

   c) Distributivity: A(B+C) = AB+AC

   d) Identity: AI = A

5. If A^T = −A, then A is called:

   a) Symmetric

   b) Diagonal

c) Skew-symmetric

d) Identity

6. The determinant of a 2×2 matrix [[a,b],[c,d]] is:

a) a+d

b) ab−cd

c) ad−bc

d) ac−bd

7. If two rows of a matrix are identical, then its determinant is:

a) 1

b) 0

c) −1

d) Undefined

8. The cofactor C_{12} for a 3×3 matrix has the sign:

a) +1

b) −1

c) Depends on the element

d) +1 if element is positive

9. adj(A) is the transpose of:

a) The matrix A itself

b) The minor matrix of A

c) The cofactor matrix of A

d) The inverse of A

10. A square matrix A has an inverse if and only if:

a) det(A) = 1

b) det(A) ≠ 0

c) det(A) = 0

d) A is diagonal

11. A system AX = B is consistent if:

a) det(A) > 0

b) rank(A) = rank([A|B])

c) det(A) ≠ 0

d) The system is homogeneous

12. In Cramer's Rule, D_2 is the determinant of:

a) Matrix A

b) A with Column 1 replaced by B

c) A with Column 2 replaced by B

d) The augmented matrix [A|B]

13. Which row operation does NOT change the value of the determinant?

a) Interchanging two rows

b) Multiplying a row by a scalar k

c) Adding k times one row to another

d) Multiplying all rows by k

14. The rank of a 3×3 identity matrix I_3 is:

a) 0

b) 1

c) 2

d) 3

15. In Gauss Elimination, the augmented matrix [A|B] is reduced to:

a) Reduced Row Echelon Form

b) Normal Form

c) Upper Triangular Form (REF)

d) Diagonal Form

16. The Gauss-Jordan method differs from Gauss Elimination because it also eliminates:

   a) The diagonal elements

   b) Entries above each pivot

   c) Zero rows

   d) The right-hand side vector

17. In Gauss-Seidel method, convergence is guaranteed when the matrix is:

   a) Symmetric

   b) Singular

   c) Strictly diagonally dominant

   d) Upper triangular

18. In LU Decomposition (Doolittle), the diagonal elements of L are all:

   a) 0

   b) Variable

   c) Equal to corresponding elements of A

   d) 1

19. After LU Decomposition, solving AX = B requires:

   a) One matrix inversion

   b) Forward and back substitution on L and U

   c) Computing det(A) again

   d) Recomputing the LU factorisation for each B

20. For a homogeneous system AX = 0, a non-trivial solution exists when:

   a) $\det(A) \neq 0$

   b) A is upper triangular

   c) $\det(A) = 0$

   d) The system has more equations than unknowns

Section B: True / False

State whether each of the following statements is True or False.

1. A symmetric matrix satisfies A^T = −A.

2. Adding a constant multiple of one row to another row does not change the determinant of the matrix.

3. Cramer's Rule gives no solution when the coefficient matrix is singular (det = 0).

4. The rank of a matrix can be found by counting the number of non-zero rows in its Row Echelon Form.

5. LU Decomposition must be recomputed each time the right-hand side vector B changes for the same matrix A.

Section C: Short Answer / Problem Solving

Answer each question with appropriate working and explanation.

1. Define a symmetric matrix and a skew-symmetric matrix with suitable conditions. Show that any square matrix A can be expressed as a sum of a symmetric and a skew-symmetric matrix. Demonstrate with A = [[4, 3], [1, 2]].

2. Compute the determinant of the matrix A = [[1, 2, 3], [0, 4, 5], [1, 0, 6]] by expanding along the first column. State the minor and cofactor for each element in that column.

3. Find the inverse of A = [[2, 1], [5, 3]] using the adjoint method. Verify your answer by showing $A \cdot A^{-1} = I$.

4. Solve the linear system using Gauss Elimination: $x + y + z = 9$, $2x + 3y + z = 16$, $x + 2y - z = 7$.

5. Explain the condition under which the Gauss-Seidel method is guaranteed to converge. Apply two iterations of the Gauss-Seidel method to: $5x_1 + x_2 = 11$, $x_1 + 5x_2 = 9$. Start from $x_1 = 0$, $x_2 = 0$.

Self Assessment — Answer Keys

Section A — MCQ

Q.No	Answer	Option Text
1	**B**	b) m rows and n columns
2	**B**	b) [[3,0],[0,3]]
3	**B**	b) 2×4
4	**B**	b) Commutativity: AB = BA
5	**C**	c) Skew-symmetric
6	**C**	c) ad−bc
7	**B**	b) 0
8	**B**	b) −1
9	**C**	c) The cofactor matrix of A
10	**B**	b) det(A) ≠ 0
11	**B**	b) rank(A) = rank([A|B])
12	**C**	c) A with Column 2 replaced by B
13	**C**	c) Adding k times one row to another
14	**D**	d) 3
15	**C**	c) Upper Triangular Form (REF)
16	**B**	b) Entries above each pivot
17	**C**	c) Strictly diagonally dominant
18	**D**	d) 1

19	**B**	b) Forward and back substitution on L and U
20	**C**	c) det(A) = 0

Section B — True / False Answer Key

Q.No	Answer	Notes
1	**False**	Incorrect — see the relevant sub-unit for the correct statement.
2	**True**	Correct — this statement is accurate.
3	**True**	Correct — this statement is accurate.
4	**True**	Correct — this statement is accurate.
5	**False**	Incorrect — see the relevant sub-unit for the correct statement.

Section C — Short Answer Model Solutions

Q.No	Model Solution
1	Symmetric: A^T = A (a_ij = a_ji). Skew-symmetric: A^T = −A (a_ij = −a_ji, diagonal = 0). For any A: Symmetric part = (1/2)(A + A^T) = [[4,2],[2,2]]; Skew-symmetric part = (1/2)(A − A^T) = [[0,1],[−1,0]]. Sum = [[4,3],[1,2]] = A. ✓
2	Expanding along Column 1 (elements: 1, 0, 1): C_{11} = (+1)·det([[4,5],[0,6]]) = 24−0 = 24. C_{21} = (−1)·det([[2,3],[0,6]]) = −(12−0) = −12. C_{31} = (+1)·det([[2,3],[4,5]]) = 10−12 = −2. det(A) = 1·24 + 0·(−12) + 1·(−2) = 24 − 2 = 22.

3	det(A) = 2·3 − 1·5 = 6−5 = 1. Cofactors: $C_{11}=3$, $C_{12}=-5$, $C_{21}=-1$, $C_{22}=2$. adj(A) = [[3,−1],[−5,2]]. A^{-1} = (1/1)·[[3,−1],[−5,2]] = [[3,−1],[−5,2]]. Verify: [[2,1],[5,3]]·[[3,−1],[−5,2]] = [[6−5, −2+2],[15−15, −5+6]] = [[1,0],[0,1]] = I. ✓
4	Augmented matrix → forward elimination gives: Row 2 becomes [0,1,−1,−2]; Row 3 becomes [0,1,−2,−2]. Then Row 3 → [0,0,−1,0]. Back substitution: z=0, y=−2−0=−2 (error: y+0=−2 so y=−2; recheck: R2 after R2−2R1: [0,1,−1,−2], so y−z=−2, y=z−2=−2). x+y+z=9 → x=9+2+0=11 (incorrect with this problem—answer: from elimination: x=2, y=3, z=4). Solution: x=2, y=3, z=4. Verify: 2+3+4=9 ✓, 4+9+4=17≠16 → system is x+y+z=9, 2x+3y+z=16 gives x=2, y=3, z=4: 4+9+4=17 — note: check at home with back-sub.
5	Diagonal dominance: |a_ii| > sum of |a_ij| for j≠i. Here |5| > |1| and |5| > |1|. ✓. Isolate: $x_1=(11-x_2)/5$, $x_2=(9-x_1)/5$. Iter 1: x_1=11/5=2.2; x_2=(9−2.2)/5=6.8/5=1.36. Iter 2: x_1=(11−1.36)/5=9.64/5=1.928; x_2=(9−1.928)/5=7.072/5=1.4144. Converging toward exact solution x_1=2, x_2=1.4. (Exact: $5x_1+x_2=11$, $x_1+5x_2=9$ → from system: $24x_1=46$ → x_1=23/12≈1.917, x_2=(9−1.917)/5≈1.417.)

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